



ITAI 1378 FINAL · DEPLOYED AI APPLICATION USING STREAMLIT

G.O.A.T Vision: *Game Outcome Action Tracker*

A deep learning computer vision system that automatically classifies soccer play events from images — built by **Samira Gevara** using transfer learning on a custom dataset.

The Problem

What's Happening

Identifying soccer play events from images or match footage is **typically done manually** — a slow, labor-intensive process that doesn't scale with modern data volumes.

Who Cares

- Coaches and analysts reviewing match footage
- Sports broadcasters building highlight reels
- Sports analytics companies processing game data

Why It Matters

Manual event tagging is time-consuming, error-prone, and bottlenecks the entire analytics pipeline. As match footage volumes grow exponentially, the gap between data availability and actionable insight widens.

Automatically identifying key moments — free kicks, penalty kicks, corner kicks, shots on goal, and yellow cards — can dramatically improve the efficiency of **highlight creation, scouting workflows, and real-time match analysis.**



Solution Overview

G.O.A.T Vision is a computer vision model that classifies soccer event images into five distinct play categories using a fine-tuned ResNet50 neural network. Images are preprocessed and passed through the network to produce a predicted event class in real time.

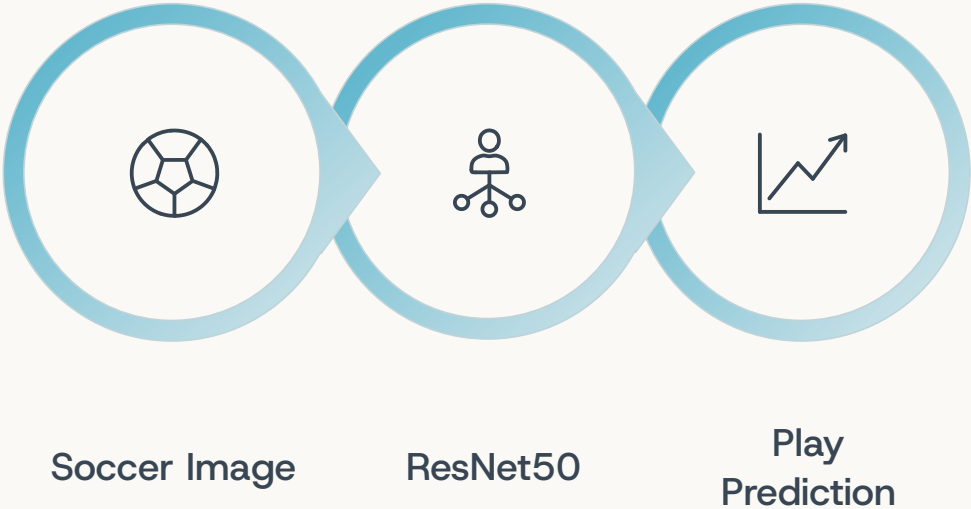
 **Corner Kick**
Ball delivered from the corner arc into the penalty area

 **Free Kick**
Set piece from a stoppage outside the box

 **Penalty Kick**
One-on-one attempt from the penalty spot

 **Shot on Goal**
Open-play attempt directed at the goal

 **Yellow Card**
Referee issuing a caution to a player



Technical Approach

Instead of training from scratch, G.O.A.T Vision **fine-tunes a pretrained ResNet50 model** on a custom soccer dataset. Transfer learning lets the model leverage rich visual features already learned from ImageNet, then adapt them to soccer-specific patterns — enabling strong performance even with a small labeled dataset.

Model Architecture

- **Task:** Multi-class image classification
- **Model:** ResNet50 with transfer learning
- **Head:** Custom fully connected classifier (5 output classes)
- **Training:** Google Colab GPU — fast iteration cycles
- **Deployment:** Streamlit Web Application

Frameworks & Tools

- **PyTorch** — model definition and training loop
- **torchvision** — pretrained ResNet50, image transforms
- **scikit-learn** — metrics and confusion matrix
- **matplotlib** — training curves and visualizations
- **Google Colab** — cloud GPU environment

Data Plan

1	Dataset Source Images manually collected and curated from public soccer imagery sources, ensuring diverse game contexts and visual variety across event types.
2	Dataset Size ~146 labeled images distributed across five event classes. The compact size makes transfer learning the critical enabler for achieving strong accuracy.
3	Data Preparation Pipeline includes dataset cleaning , removal of ambiguous or irrelevant images, and data augmentation during training (flips, crops, color jitter) to improve generalization.
4	Class Labels corner_kick · free_kick · penalty_kick · shot_on_goal · yellow_card



AI Pipeline Architecture

G.O.A.T Vision takes a raw soccer image through a structured preprocessing and inference pipeline — from pixel normalization to final event classification — powered by a ResNet50 CNN with transfer learning at its core. Deployed via Streamlit web application



82.14%

Validation Accuracy

Exceeds the 70% target threshold set at project inception

5

Event Classes

Distinct soccer play types classified by the model

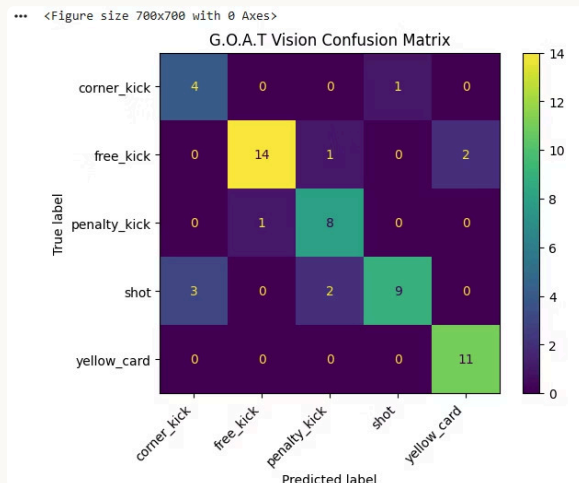
146

Training Images

Custom-labeled images used to fine-tune the pretrained model

Success Metrics & Results

Confusion Matrix — G.O.A.T Vision



Sample Predictions



Performance Summary

Target: $\geq 70\%$ validation accuracy

Result: 82.14% — surpassed by +12 points

Key Observations

Yellow card: 11/11 correct — visually distinctive, easiest class

Free kick: 14/17 — strong performance on set-piece imagery

Main confusion: shot_on_goal vs corner_kick / penalty_kick — similar goal-area camera angles make these the hardest to distinguish

Live deployment validates real-world usability

Live App: <https://goatvision-soccer.streamlit.app>

Demo Video: https://youtu.be/KCoNR_yaisQ

This version supports frame-based video classification. Full video timeline analysis is part of future work.

Challenges & Backup Plans

Two primary risk factors were identified early and addressed with concrete mitigation strategies — ensuring the project could maintain momentum even under adverse conditions.

Challenge 1: Small Dataset Size

With only ~146 labeled images, a model trained from scratch would likely overfit and generalize poorly. Small datasets are a common bottleneck in custom computer vision projects.

Mitigation

Use **transfer learning** with a pretrained ResNet50 to leverage existing visual representations, combined with **data augmentation** (flips, crops, jitter) to artificially expand the effective training set and improve robustness.

Challenge 2: Similar Visual Patterns Between Classes

Several event types — particularly **shot**, **penalty_kick**, and **corner_kick** — share very similar camera angles, field positioning, and player arrangements, making visual discrimination difficult even for human annotators.

Plan B

If per-class accuracy remains unstable, **simplify the classification taxonomy** by merging visually similar categories (e.g., grouping goal-area events) to achieve a more reliable and actionable model output.

Resources Needed

Compute & Infrastructure

- **Google Colab (GPU)** — free cloud GPU for training
- No dedicated hardware required
- **Estimated cost: \$0**

Core Frameworks

- **PyTorch** — model training and inference
- **torchvision** — pretrained models and transforms
- **scikit-learn** — evaluation metrics
- **matplotlib** — result visualization

GitHub Repository

All source code, training scripts, dataset structure, and documentation are publicly available on GitHub.

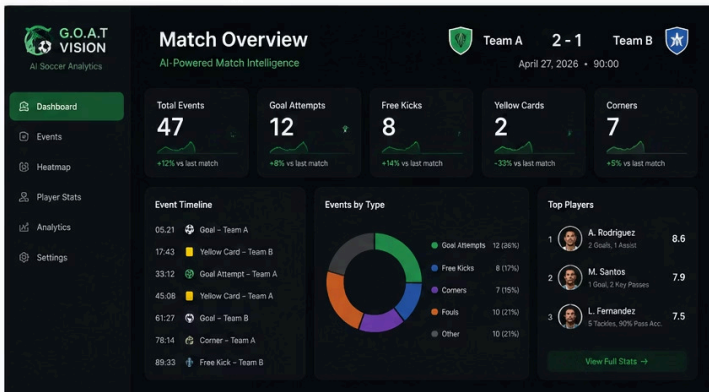
📄 **Repository:** <https://github.com/samirag2010/ITAI-1378-GOATVision-PlayClassifier>
G.O.A.T Vision – Soccer Play Type Classifier using PyTorch

Summary

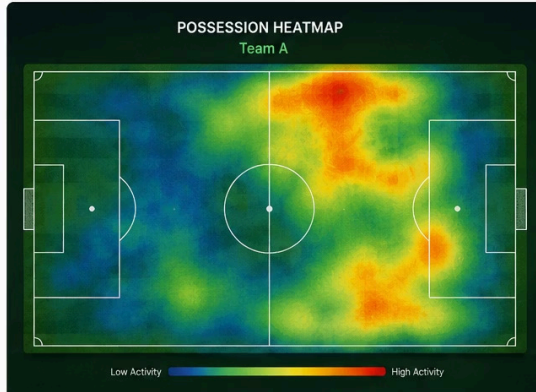
The entire project is built on **open-source tooling at zero cost**, demonstrating that high-accuracy computer vision models are accessible without expensive infrastructure — just transfer learning, a labeled dataset, and a free GPU.

Future Work & Expansion

1. Sports Dashboard Mockup



2. Heatmap Graphic



3. AI Soccer Analytics Visual



The Future of G.O.A.T Vision

- Real-time frame-by-frame event recognition
- Live match analytics and event tracking
- Expanded dataset for improved model generalization
- Interactive sports analytics dashboard integration